Comparative Study and Implementation of Generative AI Models

Assignment No. 2: AI-Assisted Comparative Study and Implementation of Generative AI Models | AIML, Batch A2 | CO3 and CO4

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1. Introduction, Aim and Objectives

Generative AI models learn the structure of data so that they can rebuild or create content. For images, two goals must be kept apart: **reconstruction**, where a model rebuilds a given input from a compressed code, and **generation**, where it produces a new image from random noise or a text prompt. This study compares an Autoencoder (AE), a Variational Autoencoder (VAE), a Deep Convolutional GAN (DCGAN) and a pretrained Stable Diffusion model.

Aim: to compare the working principles, architectures, strengths and limitations of the four models and to analyse their reconstructed and generated outputs from a practical Colab experiment. **Objectives:** (i) run AE and VAE reconstruction (CO3); (ii) train a compact DCGAN and inspect its samples (CO4); (iii) generate images with pretrained Stable Diffusion (CO4); (iv) compare the outputs without overstating the results.

2. Models Studied: Comparative Table

Aspect	Autoencoder (AE)	Variational Autoencoder (VAE)	DCGAN	Stable Diffusion (pretrained)
Primary purpose and output	Image reconstruction. Output: a reconstructed copy of the input image.	Image reconstruction and generation by sampling the latent space. Output: reconstructed or newly sampled images.	Image generation from random noise. Output: new synthetic images.	Text-to-image generation. Output: an image matching the text prompt.
Working principle	An encoder compresses the image to a latent vector; a decoder rebuilds the image. Training minimises reconstruction error [1].	The encoder predicts a mean and variance; z is sampled with the reparameterisation trick. Loss = reconstruction term + KL divergence, which regularises the latent space [2].	A generator maps noise to images while a discriminator separates real from fake; the two are trained adversarially [3], [4].	A pretrained latent diffusion model removes noise step by step from a latent, guided by the encoded text prompt, then decodes it to an image [5], [6].
Architecture (as in the notebook)	Fully connected: 784→256→64→32 (latent), then 32→64→256→784; ReLU, Sigmoid output.	Fully connected: 784→256, two heads for mean and log-variance (16 values each), then 16→256→784; Sigmoid output.	Generator: 3 transposed-convolution layers (BatchNorm, ReLU, Tanh). Discriminator: 2 convolution layers plus a linear layer (LeakyReLU, Sigmoid).	Pretrained pipeline: text encoder, U-Net denoiser and autoencoder decoder, loaded from Hugging Face [8].
Training or inference here	Trained 10 epochs on 12,000 MNIST images; Adam, lr 1e-3; MSE loss.	Trained 10 epochs on the same subset; Adam, lr 1e-3; binary cross-entropy + KL loss.	Trained 10 epochs on the same subset; Adam, lr 2e-4; noise size 64; adversarial BCE loss.	No training. Inference only: 4 prompts, 20 steps, guidance scale 7.0, 512 × 512.
Advantages	Simple and fast to train; compact representation.	Smooth probabilistic latent space; can generate new samples.	Can give sharp-edged samples; quick to sample once trained.	Detailed, realistic, text-controllable images without our own training.
Limitations	Blurring and loss of detail; latent space not designed for sampling.	Blurry outputs; random samples can be unclear.	Unstable training, artefacts, mode collapse; short training gives weak digits.	Heavy compute; sensitive to prompt wording; model access and licence rules; artefacts possible.
Applications	Compression, denoising, anomaly detection.	Representation learning, data synthesis, latent interpolation.	Data augmentation, image and texture synthesis.	Concept art, design drafts, creative content.
Observed outputs (this study)	Digit shapes kept but strokes blurred; one 4 drifted towards a 9.	Similar blur; both 4s recognisable, the 2 more distorted; random samples mostly malformed.	Digit-like strokes with many broken or odd shapes; a few recognisable digits.	Sharp, realistic nature scenes; minor artefacts (stray branch piece, garbled marks in one corner).

Table 1 Comparison of the four models. Architecture and training details are taken from the executed notebook.

3. Implementation and Experimental Setup

All experiments ran in a Google Colab notebook (Python 3.13.16, PyTorch 2.11.0+cu130, torchvision 0.26.0+cu130) on an NVIDIA Tesla T4 GPU, with random seed 42. AE, VAE and DCGAN were trained on the first 12,000 MNIST training images (batch size 128, 10 epochs each) and evaluated on the first 2,000 MNIST test images. Stable Diffusion (model ID stable-diffusion-v1-5/stable-diffusion-v1-5) was loaded as a **pretrained** pipeline through the diffusers library and used only for inference; the notebook disabled its safety checker.

Scope note: AE, VAE and DCGAN work on MNIST handwritten digits, whereas Stable Diffusion generates nature and wildlife images from text prompts. The study therefore compares model capabilities and output types; it is not a controlled, same-domain image-quality benchmark.

4. Results and Observations

Model	Epoch 1 loss	Epoch 10 loss	Training loss definition	Test MSE (first 8 test images)
AE	0.089489	0.022156	Mean squared error per pixel	0.018874
VAE	248.556259	119.692467	Per-image sum of binary cross-entropy + KL divergence	0.018648

Table 2 Measured values from the notebook. The AE loss (MSE) and VAE loss (cross-entropy + KL) have different definitions and scales, so the training losses are not compared with each other.

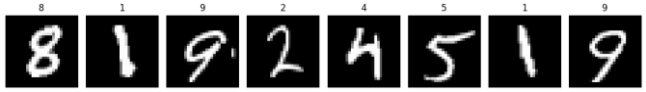


Fig. 1 Sample MNIST training digits (dataset preview).

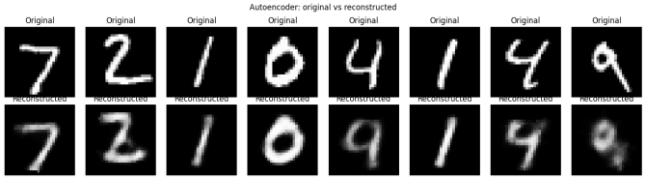


Fig. 2 AE: original (top) vs reconstructed (bottom).

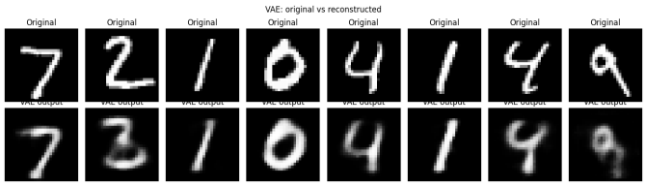


Fig. 3 VAE: original (top) vs reconstructed (bottom).

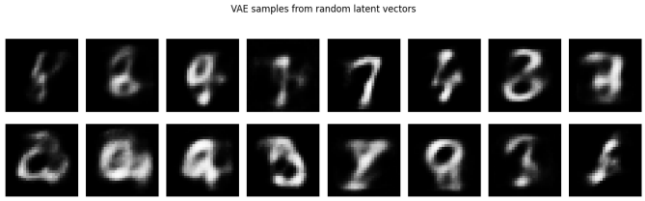


Fig. 4 VAE samples decoded from random latent vectors.

AE and VAE (Figs. 2–4). Both models used the same eight test digits. Both keep the overall shapes of the 7, 1, 0 and 1 but blur the strokes. The AE turned one 4 towards a 9 and smudged the final 9. The VAE kept both 4s recognisable but distorted the 2 more than the AE. The test MSE values are 0.018874 (AE) and 0.018648 (VAE). This 0.000226 gap comes from only eight images, the VAE reconstruction uses a randomly sampled latent vector, and the two models differ in size and loss, so it does not show that one model is better. VAE samples decoded from random latent vectors (Fig. 4) are mostly malformed, with only a few digit-like shapes.

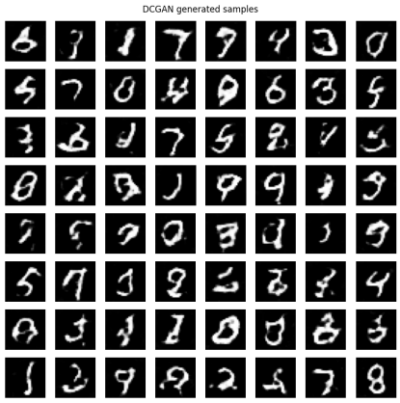


Fig. 5 DCGAN: 64 generated samples after 10 epochs.



Fig. 6 Stable Diffusion: tiger, kingfisher, mountain sunrise, wildflowers (left to right, top to bottom).

DCGAN (Fig. 5). After 10 epochs the generator produces digit-like strokes in varied shapes, but many samples are broken or ambiguous; a few look like a 0, 7, 8 or 9. Mean epoch losses moved from 0.6166 (discriminator) and 2.1622 (generator) in epoch 1 to 0.6663 and 1.6046 in epoch 10; these are adversarial losses, not image-quality scores.

Stable Diffusion (Fig. 6). The tiger image shows clear fur and stripes on a forest path. The kingfisher is detailed, but the river is not clearly visible and a stray branch piece floats above it. The mountain scene has layered mist and a warm sunrise glow without a distinct sun. The wildflower image has soft purple flowers but no clear signs of rainfall, and faint garbled text-like marks appear in the bottom-right corner. No numerical quality score was recorded for DCGAN or Stable Diffusion.

5. Comparative Discussion

The four models answer different questions. AE and VAE are judged by how closely the output matches a given input, so reconstruction error is meaningful. A VAE additionally shapes its latent space so that random sampling is possible, but this regularisation tends to give blurry output. DCGAN has no input image to match; the generator is judged only by whether the discriminator accepts its samples, which gives sharper strokes but less stable training. Stable Diffusion adds text conditioning, so its images are judged by realism and by how well they follow the prompt, which is why the notebook records no MSE for it.

6. Conclusion

Reconstruction models (AE, VAE) are simple and suit compression and denoising but blur detail. The DCGAN can create new digits but needs more training for clean results. The pretrained diffusion model produced the most detailed and realistic images and responds to text, at a higher computational cost and with minor artefacts. No model is best for every task: the right choice depends on whether the goal is faithful reconstruction, sampling, or controllable generation. The study is limited by small data subsets, 10-epoch training, a single run, and different data across the models.

7. AI Usage Documentation and Group Reflection

- **AI tools used:** Claude (Anthropic) drafted this report and the presentation from the assignment PDF, the executed notebook and a design reference. [Group to add any other AI tools used for the notebook.]
- **Important prompt (summary):** asked Claude to read the assignment PDF and executed notebook, never invent results, and produce a 2–3 page DOCX report and a 5-slide editable PPTX matching a reference design. [Group to add other prompts.]
- **Verification:** metrics, settings and prompts were checked against the notebook's printed outputs; all figures are the notebook's own output images; observations were written after viewing each figure; model descriptions follow the references below.
- **Key learning:** reconstruction and generation are different goals; a lower reconstruction MSE alone cannot rank unlike models; pretrained models allow high-quality generation without training from scratch.

Contribution of each group member

Group member	Contribution (2–3 lines each)
Parth Kitchloo (202401110014)	Set up and ran the Colab notebook on the T4 GPU, and trained the AE, VAE and DCGAN on the MNIST subset. He recorded the epoch losses and test MSE values and saved the reconstruction and generated-image figures. He also led the AE versus VAE reconstruction analysis.
Ragav Sharma (202401110012)	Ran the Stable Diffusion pipeline with the four wildlife prompts and analysed the generated images. He also designed the presentation in the reference template style, and checked the report and slides against the notebook outputs before final submission.
Gufran Shah (202401110011)	Researched and wrote the working-principle, advantages, limitations and applications entries for the comparison table. He also collected and checked the references.
Sandeep Kotwal (202401110052)	Compared the DCGAN and Stable Diffusion outputs, proofread the report and slides for spelling and consistency, and helped prepare the 4–5 minute presentation.

References

[1] G. E. Hinton and R. R. Salakhutdinov, "Reducing the dimensionality of data with neural networks," Science, vol. 313, no. 5786, pp. 504–507, 2006.

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[4] A. Radford, L. Metz, and S. Chintala, "Unsupervised representation learning with deep convolutional generative adversarial networks," in Proc. ICLR, 2016. arXiv:1511.06434.

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[7] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," Proc. IEEE, vol. 86, no. 11, pp. 2278–2324, 1998. (MNIST dataset)

[8] Hugging Face, "stable-diffusion-v1-5/stable-diffusion-v1-5" model page. https://huggingface.co/stable-diffusion-v1-5/stable-diffusion-v1-5